

1 AI-Powered Drone Capable Of GPS-Free Navigation

2 "Why Buy A New Kit Every Time When One Can Do It All?"

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1 AI-POWERED DRONE CAPABLE OF GPS-FREE NAVIGATION

VECROS, a Bengaluru-based startup, builds AI-powered, GPS-free autonomous drones for defence, construction, and warehouse applications with advanced vision systems.



Athera drone

Drones that see, think, and react like fighter pilots, making split-second decisions on their own, are no longer futuristic. Bengaluru-based VECROS Technologies Pvt Ltd, founded in 2021 by Besta Prem Sai and Rajeshree Deotalu, has made them real.

Besta Prem Sai outlines the company vision, highlighting two core challenges in the drone industry: dependence on GPS and the requirement for skilled pilots to manually operate each drone. Traditional drones rely heavily on GPS for navigation, making them highly susceptible to GPS jamming, an issue that can instantly disable a drone and poses a serious risk in electronic warfare, particularly within defence operations.

The company addresses this vulnerability by removing the need for GPS entirely. Instead, it uses

advanced vision systems and AI models that allow drones to interpret their surroundings and determine their location through visual landmarks. By 'seeing' and understanding the terrain, the drones can navigate independently of GPS. This capability for GPS-denied navigation is a key differentiator, especially in military environments where GPS interference is an escalating threat.

"The second major challenge lies in the current operational model, which demands one pilot per drone," Besta explains. "Our goal is to transform this approach by creating a system that enables a single pilot to manage multiple drones, or even allows the drones to operate autonomously using intelligent software."

VECROS has introduced two flagship products, Athera and Jasper. Athera is described as India's

first AI-powered drone capable of operating without GPS. It has approximately 10 onboard cameras and performs real-time AI processing, enabling autonomous operation while ensuring data privacy. It has a 30-minute flight time and a dual-swappable battery system that allows rapid redeployment within five minutes.

Jasper, a smaller variant built with similar AI capabilities, is designed for confined spaces and is used in situations where compact and agile drones are required. While both products are operational, they are currently available for enterprise use rather than consumer markets.

The core value of VECROS lies in its proprietary software and custom electronics, particularly the onboard GPU-based processing system that allows drones to process AI tasks locally. The technology stack includes a custom-built motherboard (JetCore) designed to handle power and communications efficiently.

VECROS targets three primary industries: warehouse management, construction, and defence. It currently works closely with these sectors to deploy autonomous drones for applications such as 3D scanning, elevation mapping, and ISR (intelligence, surveillance, and reconnaissance) missions in defence. These use cases benefit